

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/226712347>

Does the Ecoregion Approach Support the Typological Demands of the Eu 'water Framework Directive'?

Article in *Hydrobiologia* · March 2004

DOI: 10.1023/B:HYDR.0000025256.73580.3e

CITATIONS

109

READS

785

4 authors:



Otto Moog

BOKU University

286 PUBLICATIONS 5,950 CITATIONS

[SEE PROFILE](#)



Astrid Schmidt-Kloiber

BOKU University

91 PUBLICATIONS 3,054 CITATIONS

[SEE PROFILE](#)



Thomas Ofenböck

Vienna City Administration

46 PUBLICATIONS 1,462 CITATIONS

[SEE PROFILE](#)



Jeroen Gerritsen

Tetra Tech

80 PUBLICATIONS 9,708 CITATIONS

[SEE PROFILE](#)



Does the ecoregion approach support the typological demands of the EU 'Water Framework Directive'?

Otto Moog¹, Astrid Schmidt-Kloiber¹, Thomas Ofenböck¹ & Jeroen Gerritsen²

¹University of Agricultural Sciences, Institute for Water Provision, Aquatic Ecology and Waste Management, Department of Hydrobiology, Fisheries Management and Aquaculture, Max-Emanuel-Strasse 17, A-1180 Vienna, Austria

Tel.: +43-1-47654-5200. Fax: +43-1-47654-5217;

E-mail: otto.moog@boku.ac.at; astrid.schmidt-kloiber@boku.ac.at; thomas.ofenboeck@boku.ac.at

²Tetra Tech, Inc., 10045 Red Run Blvd., Owings Mills, MD 21117, U.S.A.

Jeroen.Gerritsen@tetratech.com

Key words: Austria, benthic macroinvertebrates, bioregion, ecoregion, running waters, water framework directive

Abstract

The new European assessment system for freshwater ecosystems is based on reference conditions. As the ecological status of rivers must be determined with respect to the near-natural reference conditions focus is given to establish a typological framework. A basic part of this typology is the ecoregion approach. Multivariate analysis of the benthic invertebrate assemblages confirmed the grouping of Austrian fluvial landscapes into four of the European aquatic ecoregions: Alps, Central Highlands, Hungarian Lowlands and the Dinaric Western Balkan. The benthic assemblages are more homogeneous within an ecoregion than among different ecoregions. Analyzing sites from adjacent areas in Italy, Germany and the Czech Republic clearly indicated that the ecoregion concept fulfils the goal of independence from political borders and that this concept can be used in Europe for implementing the Water Framework Directive. Nevertheless, considering the detailed structure of central European landscapes, the ecoregions are too large and variable to provide a framework for regional management based on ecological data. With respect to the variety of the central European river faunas a further division into bioregions was seen as necessary. Combining *a priori* derived types of Austrian riverine-landscape-types and major surface-water-types with the results of the multivariate analysis (groups of similar benthic assemblages) led to a description of 15 'Austrian Fluvial Bioregions'. Fluvial Bioregions are very strong descriptors of the faunal assemblages; their determinative character is stronger than the ecological quality of a site. Within the bioregions, the natural variability of several biological measures (metrics, indexes) is low. Consequently, the discriminatory power of the measures for evaluating ecological status is increased by using the bioregional classification.

Introduction

Since December 2000, European water management is ruled by a new common legislative construct establishing a 'Framework for Community Action in the Field of Water Policy' (European Commission, 2000). This European Water Framework Directive (WFD) provides the first comprehensive general concept for water management and protection of freshwater ecosystems in Europe. The environmental objective of the directive is to maintain or achieve at least 'good

ecological status' for all surface waters by 2015, to be realized through programs of measures as defined in River Basin Management Plans (Koller-Kreimel & Chovanec, 1999). The benefit of the WFD for the European Union partner countries is that rivers are seen and managed as ecosystems by a single legislative framework that sets uniform standards in water policy throughout the European Union. The WFD acknowledges for the first time the importance of aquatic biota in assessing the quality of European fresh and

coastal waters (Logan & Furse, 2002). In setting ecological targets for surface waters, the Commission has recognized the need for an integrated approach to manage three of the key components of aquatic habitats: water quality, water quantity, and physical structure (Logan, 2001). This implies that the ecological status of water bodies must be determined with respect to the near-natural reference conditions, which represent a 'high ecological status'. Significantly, the determination of the 'good ecological status' is based on characterizing reference conditions. The WFD approach of using reference conditions to assess the ecological status of waters is thus concordant with biological assessment adopted in several countries (e.g. the U.S.: USEPA, 1996; the U.K.: Hemsley-Flint, 2000; Austria: Austrian Standards ÖNORM M 6232, 1997, Chovanec et al., 1994, 2000; Australia: Davies, 2000).

To describe reference conditions and to conduct a reference based quality assessment a typological framework is needed (Verdonschot, 2004; Nijboer et al., 2004). In the WFD water bodies are categorized as rivers, lakes, transitional or coastal waters. For each of these categories, the water bodies have to be differentiated according to type, using one of two systems (System A or B). The objective of the typology is to ensure that type-specific biological reference conditions can be determined. An essential part of the System A – typology is the ecoregion approach, together with three additional features: altitude, catchment area, and geology. System B for rivers includes both 'obligatory' and 'optional' factors; the obligatory factors are those included in System A. Whereas the factors in system A define environmental features that influence biological communities but are not themselves influenced by man, the optional habitat features in System B could be affected by human activities. By including the ecoregion approach in System A, the directive overcomes the fact that national policies and planning for biodiversity management take place at the national level, but biodiversity and ecological processes do not conform to national boundaries.

The aims of this paper are: (1) to summarize the current state of the applicability of the top-down and bottom-up ecoregion approaches (Moog et al., 2001a; Schmidt-Kloiber et al., 2001) and bioregion approaches (Moog et al., 2001b) for characterising typological features of Austrian rivers and streams and (2) to validate these (zoo)geographic typologies focusing on the benthic macroinvertebrate communities in adjacent countries.

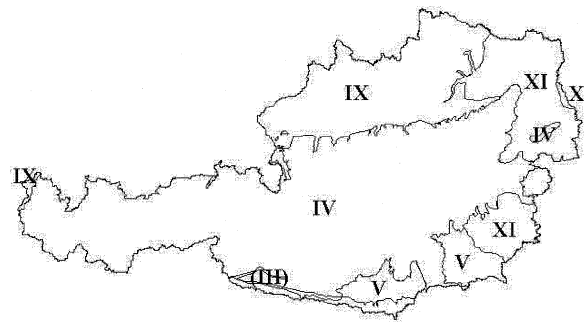


Figure 1. The ecoregions of Austria according to Moog et al. (2001a): III = Italy; IV = Alps; V = Dinaric Western Balkan; IX = Central Highlands; X = The Carpathians; XI = Hungarian Lowlands; borders of ecoregions III and X only outlined schematically.

Austrian landscape typologies – *A priori* approaches

Ecoregions

The ecoregions of Austria were adopted according to the European delineation of Illies' (1978) *Limnofauna Europaea*. Illies divides Europe into 25 regions with roughly defined boundaries. The rather coarsely defined borders of Illies' ecoregions were defined more precisely for practical use in Austria (Moog et al., 2001a), using geological and ecological features, quite similar to the delineation process in the U.S. (e.g., Omernik, 1987, 1995) (Fig. 1).

Austria (about 84 000 km²) comprises parts of 6 ecoregions. Four ecoregions are of sufficiently large area in Austria for use as *a priori* classes for aquatic community types:

Alps (ecoregion 4) – The Alps mountains, as defined from approximately the 500 to 1000 m contour, cover 63.5% of Austria. They extend from the borders of Liechtenstein and Switzerland in the west, to Italy and Slovenia in the south and to the Piedmont along the Danube in the north.

Dinaric Western Balkans (ecoregion 5) – These lowlands and basins are in the south of Austria, along the border with Slovenia covering 3.4% of the federal territory.

Hungarian Lowlands (ecoregion 11) – The Austrian part of the Hungarian Lowlands, including the Vienna Basin, is situated in northeastern Austria and extends from the Alps and the Central Highlands north and east to the borders with Slovakia and Hungary (10.5% of the federal territory).

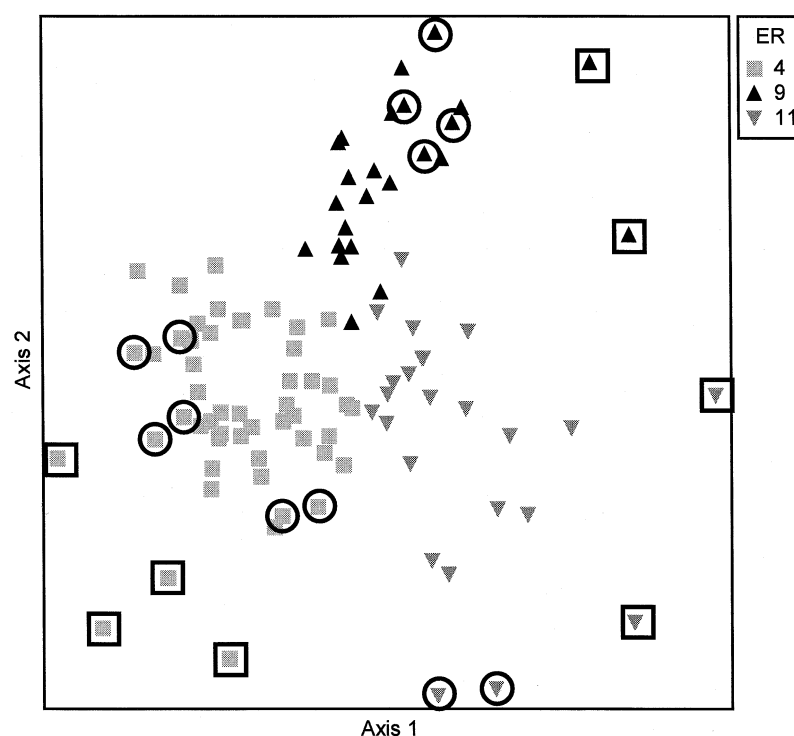


Figure 2. Scatter plot of AQEM sampling sites with ecoregions-overlay (sites of any quality, species level comparison): ER4 = Alps; ER9 = Central Highlands; ER11 = Hungarian Lowlands; circles indicate sites of poor and squares sites of bad quality classes; stress = 16.12; $r^2 = 0.44$.

Central Highlands (ecoregion 9) – In Austria the Central Highlands (22.6% of Austria) comprise two quite different sub-ecoregions, the piedmont along the Danube west of Vienna and the granite-gneiss mountains of the Bohemian Massif extending north to the borders of Germany and the Czech Republic.

The two regions of minor importance in Austria are small extensions of the *Carpathians (ecoregion 10)* and the *Italy (ecoregion 3)* (Fig. 1).

River landscape types

Given the range of elevations in Austria, and the complex geology of the Alps and Central Highlands, the ecoregions may be too coarse to provide a framework for organizing ecological data for a regional management approach. Therefore we examined whether a further division into sub-ecoregions or other typological regions was desirable, and a system of river-landscape-types was developed to obtain a more detailed partition of the ecoregions. Using an *a priori* approach, an exhaustive division of the ecoregions into 40 river-landscape-types based on features of geology, geomorphology, climate, watersheds, discharge re-

gime type, vertical vegetation zones, vegetation types and others was developed (Fink et al., 2000).

Major surface-water-types

These 40 river-landscape-types served as a foundation for establishing a set of 26 major surface-water-types (17 ‘aquatic landscape units’ and 9 ‘large Rivers’) in accordance with the methodological approach of the WFD as given in the Annex II (System A and B) (Wimmer & Chovanec, 2000; Wimmer et al., 2000). These major surface-water-types are the basis for the development of a surveillance monitoring network as required by the WFD. The 17 Aquatic Landscape Units were the *a priori* classifications that we principally tested and modified to generate aquatic bioregions for Austria.

Biological components

Under the WFD, biota are the most important component for the evaluation of ecological status. The communities investigated in rivers include: algae, macrophytes, benthic invertebrates, and fish. To describe the biological elements the following attributes have to be

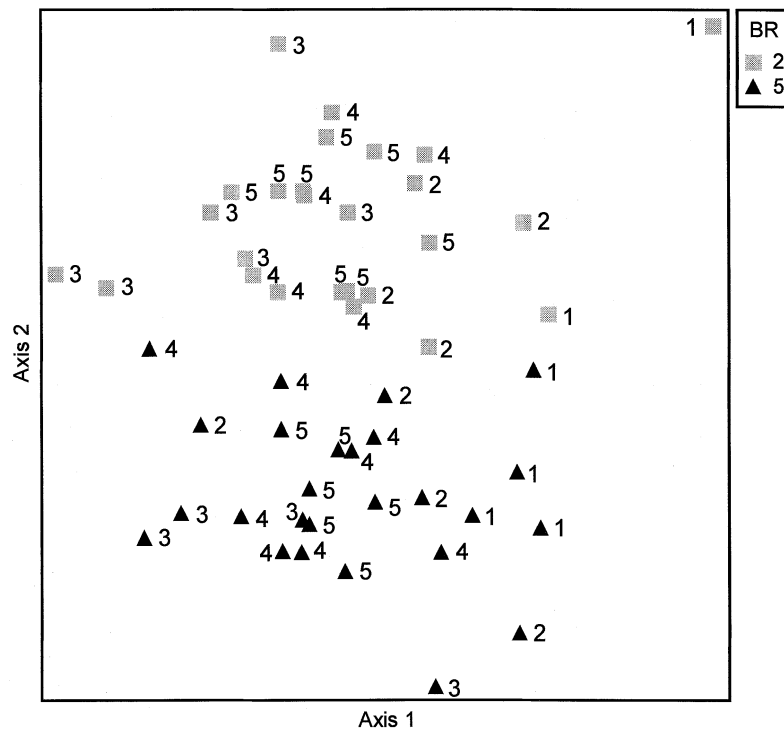


Figure 3. Scatter plot of AQEM sampling sites in the ecoregion 'Alps', the overlay indicates the bioregion: BR2 = Non-glaciated Crystalline Alps and BR5 = Limestone Foothills; numbers beneath the sites indicate the quality classes: 5 = high, 4 = good, 3 = moderate, 2 = poor, 1 = bad; stress = 11.9; $r^2 = 0.59$.

considered: taxonomic composition, abundance, the ratio of disturbance-sensitive taxa to insensitive taxa, diversity and the age structure of fish populations. Hydrology, physical habitat, and physico-chemical water quality are investigated in addition to the biological components.

To describe reference conditions a classification or typological framework is needed. A basic part of this typology is a geographic delineation that reflects the community composition of the aquatic biota (e.g., Gerritsen et al., 2000).

The objective of this research was to identify a set of biological regions for the streams of Austria, based on the major surface-water-types. The ecoregions, the riverine landscape types, and the surface water types served as three sets of *a priori* hypotheses for partitioning of benthic macroinvertebrate community types. The boundaries of the ecoregions were adjusted and tested, and modified if necessary, based on biological information and expert judgment (Moog et al., 2001a). The biological regions are defined by assemblage composition, and thus form the basis for practical biological assessment using reference conditions (Barbour et al., 1999). The expectation is that

each biological region has a predictable composition of the benthic macroinvertebrate assemblage.

Methods

The approach we took was to assemble data sets on the benthic macroinvertebrates collected by several monitoring and research entities throughout Austria, and subject these to quality assurance filters so that each analysis comprised only compatible data. Multivariate analysis of the species composition was applied to the data sets, separately and combined, to examine the affinity between macroinvertebrate assemblages and the *a priori* regionalization.

Our approach throughout was to examine the *a priori* regionalization with the species composition of least-disturbed reference sites. Classification of species composition data usually involves ordination or clustering of sites to examine classes and structure. These require a distance or similarity measure for site pairs, calculated from the taxonomic composition of the sites.

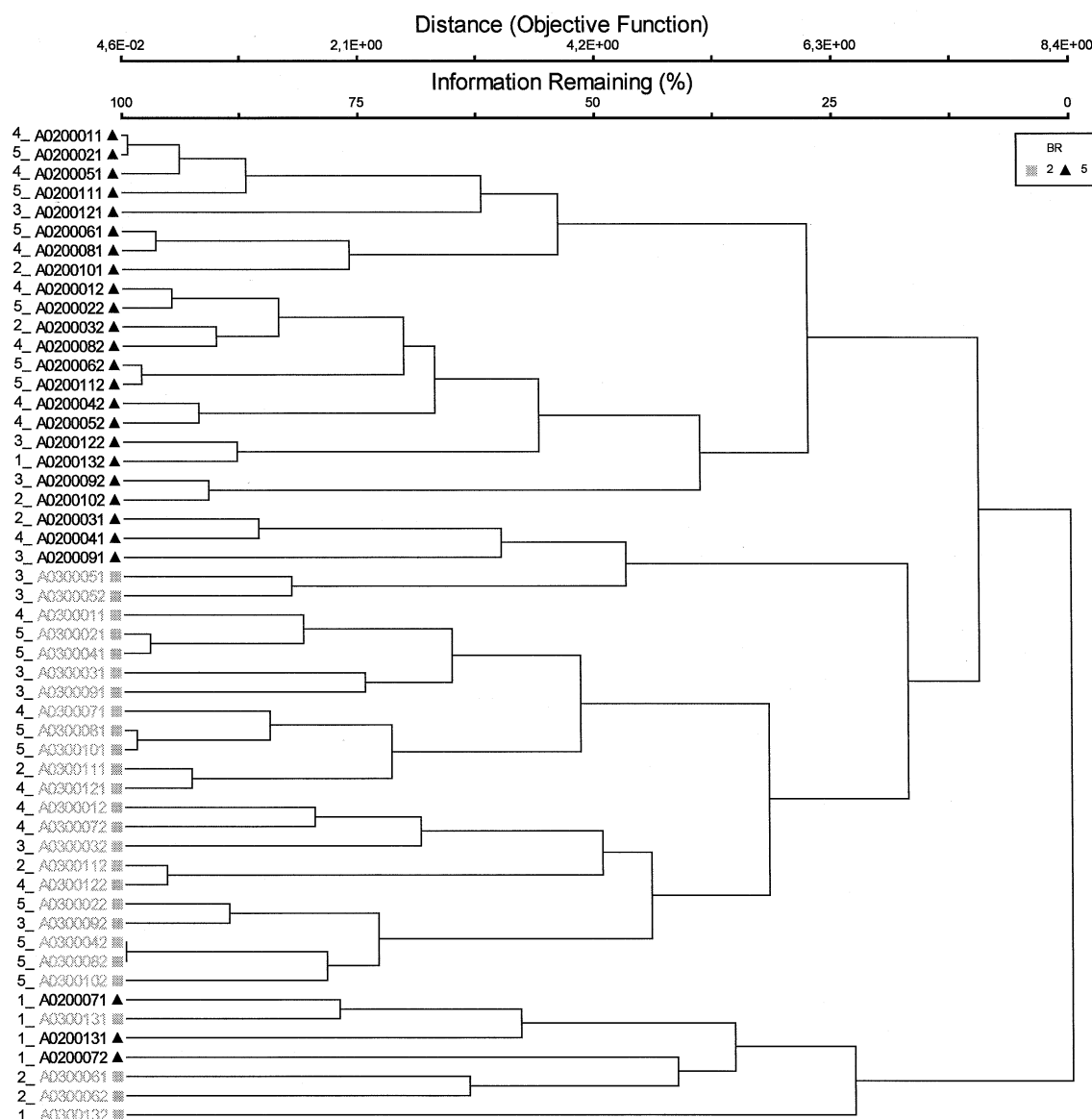


Figure 4. Cluster of AQEM sampling sites in the ecoregion 'Alps', the overlay indicates bioregion: BR2 = Non-glaciated Crystalline Alps and BR5 = Limestone Foothills; the quality classes are indicated as numbers on the left: 5 = high, 4 = good, 3 = moderate, 2 = poor, 1 = bad.

A primary requirement for a data set is taxonomic consistency, or always identifying a given taxon as the same thing (Nijboer & Verdonchot, 2000; Schmidt-Kloiber & Nijboer, 2004). The data sets we used here originated from multiple sources, and required reconciliation to ensure that taxonomic level and precision were comparable (taxonomic adjustment). In the event of taxonomic disagreement among data sets, taxa were transformed to a higher level or excluded from the taxa list. For example, if the discrepancy occurs at the species level in a given genus, then all observations of

(multiple) species within that genus can be collapsed to the genus level for data analysis. If the discrepancy concerns one or more rare taxa, they can safely be excluded from analysis.

Data sets

The analyses were based on a data set of more than 5000 sites that had been reduced to 582 sites meeting reference criteria (according to expert consensus of state and federal biologists and administrators), and

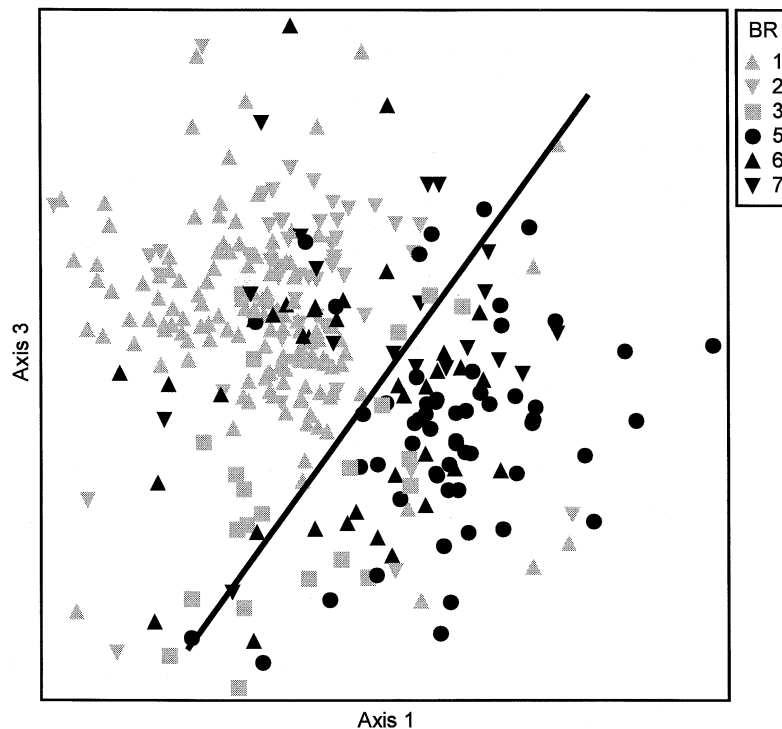


Figure 5. Scatterplot for all Austrian sampling sites with reference status, showing divisions among principal ecoregions (according to Moog et al., 2001b): Crystalline Alps (light grey; BR1-3) and Limestone Alps (dark grey; BR5-7); for numbers of bioregions (BR) see Table 1; stress = 18.57; $r^2 = 0.45$.

sufficiently high quality species and/or genus level identifications. We used genus level data for large area classification because of the non-homogeneity of the data sets that comprised samples taken throughout Austria by different agencies. We used species level data for small geographic areas if the data were of homogeneous quality.

Some data sets had reliable relative abundance information, and others had only qualitative presence/absence information. Analyses requiring relative abundances could use only the appropriate data, but analyses requiring only presence/absence data could use all the data. Data sources included: Austrian federal water monitoring agencies; state and municipal water monitoring agencies; contributed studies; dissertations; diploma theses, and other studies from the Benthic Ecology Group at BOKU-Vienna; and the EU-financed project AQEM, specifically to develop an assessment system using benthic macroinvertebrates.

Multivariate benthic community analysis

Multivariate ordination can graphically demonstrate whether the biotic species composition reflects the hy-

pothesized *a priori* classification. We used non-metric multidimensional scaling (NMS) as the ordination method.

Ordination analysis is a means of reducing the complexity of multivariate data so that it can be visualized graphically, and examined with other, more conventional, exploratory analyses. Non-metric multidimensional scaling develops an ordination from any distance or similarity matrix. The procedure is to rank the distances in the original matrix, and then attempt to display those ranks of distances in a specified number of dimensions, usually 2 or 3. In effect, NMS reproduces a map of the entities from the distances among the sites. The goodness-of-fit of the estimated distances is measured by the stress statistic (indicated in the figures). Additionally a coefficient of determination (r^2) for the correlations between ordination distances and distances in the original n -dimensional space is calculated and indicated as the cumulative value of the axis shown in the figures. The computational procedure is numerical approximation, beginning with an arbitrary configuration and reducing the stress statistic in each successive approximation. When plotted in ordination space, sites with similar

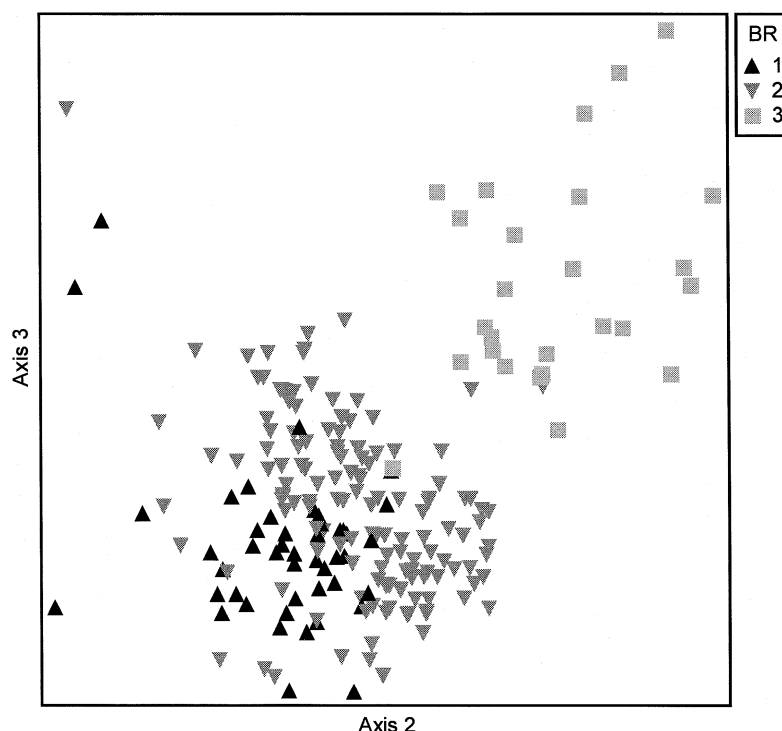


Figure 6. Scatterplot for all Austrian sampling sites with reference status from the Central Crystalline Alps indicating a division into three bioregions (according to Moog et al., 2001b): Glaciated Crystalline Alps (BR1), Non-glaciated Crystalline Alps (BR2), Ridges and Foothills of the Crystalline Alps (BR3); stress = 17.24; $r^2 = 0.51$.

species composition will be close together. Ordination plots are thus used to verify or falsify the *a priori* classification hypotheses. NMS is relatively robust for species composition data, and has been applied frequently in recent years to benthic macroinvertebrate data (e.g., Reynoldson et al., 1995; Barbour et al., 1996).

The Soerensen/Bray–Curtis coefficient was used as the distance measure. With presence/absence data only, it is equivalent to the Jaccard coefficient (Legendre & Legendre, 1998). To quantify classification strengths of alternative *a priori* classes, we followed the similarity analysis of Van Sickle (1997) and Van Sickle & Hughes (2000). The similarity analysis gives a quantitative estimate of the amount of similarity within and among groups that is explained by the classification. Because our results were quite strong, we do not discuss the similarity analysis in detail. More detailed information can be taken from Schmidt-Kloiber (2002). Finally we cross-validated the classifications with cluster analysis using the Soerensen coefficient and flexible beta (beta set to -0.25 ; Hawkins et al., 2000; Gerritsen et al., 2000). The

cluster analysis yields a quantitatively ‘best’ classification in terms of the within and among-group mean similarity. A ‘quantitative best’ classification is not necessarily the most desirable because it may not yield an explanation or a method of determining the expected class of a stressed or impaired site.

Prediction of expected type can be developed with a geographic ecoregion model, or with a discriminant analysis based on physical data (e.g., Moss et al., 1987; Hughes and Larsen, 1988; Barbour et al., 1996; Moss, 2000). Our approach was to modify and adjust the *a priori* models to develop an optimum geographic and physical classification.

NMS and cluster analyses were performed with PC-ORD 4.1 (McCune & Mefford, 1999); gemicol for mean similarity analyses MEANSIM 6.0 (Van Sickle, 1997) was used.

Results

Previous analysis had demonstrated that benthic macroinvertebrate communities could be explained by ecoregions (Schmidt-Kloiber et al., 2001): reference

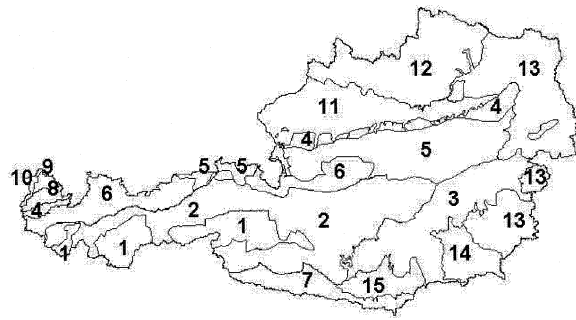


Figure 7. Map of the Fluvial Bioregions of Austria according to Moog et al. (2001b); for the legend see Table 1.

sites throughout Austria formed groups according to ecoregions, in spite of methodological differences among data sets. The strongest division when analyzing all data together (including methodological differences) was between the Alps and the other regions of Austria. When data sets were examined individually, the ecoregional separations were sharper (Schmidt-Kloiber et al., 2001).

The data set from the AQEM project was the most consistent and precise data source, consisting of 98 sites in Austria from all 5 ecological quality classes (bad to undisturbed). Candidate ecoregions could be discriminated in the AQEM data, even when all 5 quality classes were included. Figure 2 demonstrates the scatter plot with an ecoregions overlay indicating a clear grouping of sites of the Alps (ER 4), the Central Highlands (ER 9), and the Hungarian Lowlands (ER 11). Also presented in Fig. 2 are the impaired rivers sites, circles indicating poor and squares indicating bad ecological status. Most of the sites of poor and bad ecological quality were located at the edges of the bioregion scatter plots, but in the vicinity of the good quality sites of the corresponding ecoregions.

A detailed analysis of the alpine sites from the AQEM data shows the division of the Alps into two subdivisions more clearly. The projection of the ordination indicated no overlap between the stream faunas of the Limestone Foothills and the Non-glaciated Crystalline Alps (Fig. 3). Additionally the ecological status of the sites is indicated by numbers. The two regions separate distinctly on the vertical axis and the ecological status separates on the horizontal axis, but not as distinctly. Thus, the quality of a site does not greatly affect the similarity of sites within a bioregion. The 'ecological status classes' of this analysis comprised the full set of quality classes. The separation of the regions compared to ecological status, gives evi-

Table 1. The Fluvial Bioregions of Austria according to Moog et al. (2001b)

Bioregions within the ecoregion 'Alps'	
Sub-ecoregion 'Central Crystalline Alps'	
1.	Glaciated Crystalline Alps
2.	Non-glaciated Crystalline Alps
3.	Ridges and Foothills of the Crystalline Alps
Sub-ecoregion 'Limestone Alps'	
4.	Flysch (Sandstone) Mountains and Ridges
5.	Limestone Foothills
6.	Limestone Alps
7.	Southern Alps
8.	Helveticum in Vorarlberg
9.	Alpine Molasse
Bioregions within the Ecoregion 'Central Highlands'	
Sub-ecoregion 'Piedmont'	
10.	Piedmont in Vorarlberg
11.	Bavarian-Austrian Piedmont
Sub-ecoregion 'Bohemian Massif'	
12.	Granite and Gneiss Region of the Bohemian Massif
Bioregions within the ecoregion 'Hungarian Lowlands'	
13.	Eastern Ridges and Lowlands
Bioregions within the Ecoregion 'Dinaric Western Balkan'	
14.	Grazer Basin and Grabenland
15.	Southern Inner-alpine Basins

ence that the determinative character of bioregions is stronger than the ecological quality of a site.

Cluster analysis also confirmed the *a priori* division of the Alps into the two bioregions of Non-glaciated Crystalline Alps and Limestone Foothills (Fig. 4). The two bioregions, comprising 52 sites in all 5 ecological quality classes of the AQEM data set, formed two distinct clusters, with only 7 outliers that could not be placed in the clusters.

The high separation seen in these analyses can be explained by the high quality of the AQEM data. Sampling design, pre-classification of sites, field work, and laboratory processing in the AQEM project were standardized and a high taxonomic precision of identification was applied. More detailed methodological information is given by Hering et al. (2003).

To address the possibility of other subdivisions of the ecoregions, following the *a priori* hypotheses of river landscapes and surface water types, we ex-

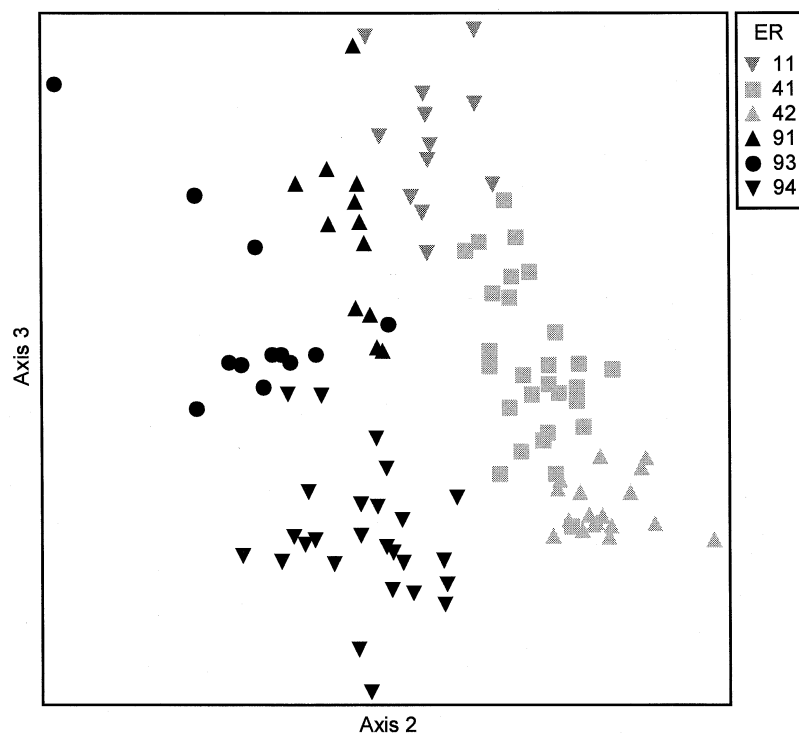


Figure 8. Scatterplot for AQEM sampling sites with reference and good status in adjacent countries in three European ecoregions: Alps (ER4), Central Highlands (ER9), Hungarian Lowlands (ER11); the last digits of the legend indicate the different countries (1 = Austria, 2 = Italy, 3 = Czech Republic, 4 = Germany); stress = 15.3; $r^2 = 0.42$.

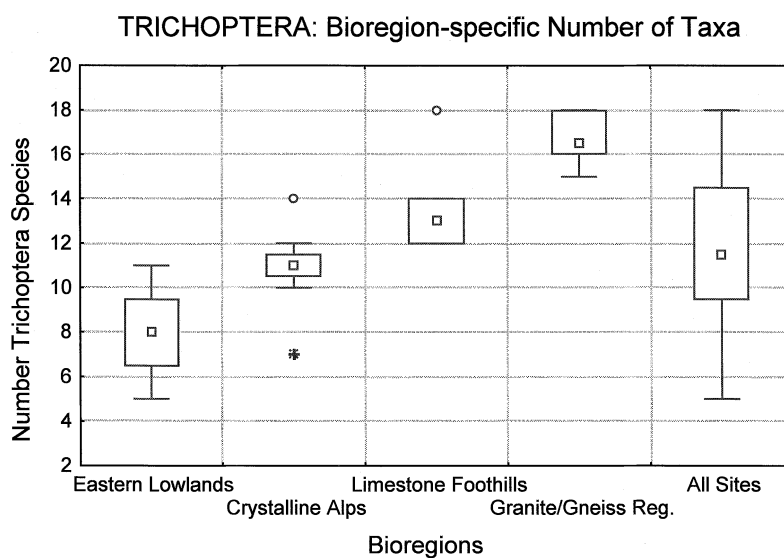


Figure 9. Number of Trichoptera species in rivers of four different bioregions of Austria investigated in the AQEM project.

amined data from individual ecoregions, but with all data sources combined. For example, Fig. 5 shows Alps reference sites from all data sets ($n = 315$). The assemblages of the limestone Alps are clearly distinct

from the crystalline Alps. This analysis with independent data confirms the separation into two subgroups, Crystalline Alps and Limestone Alps.

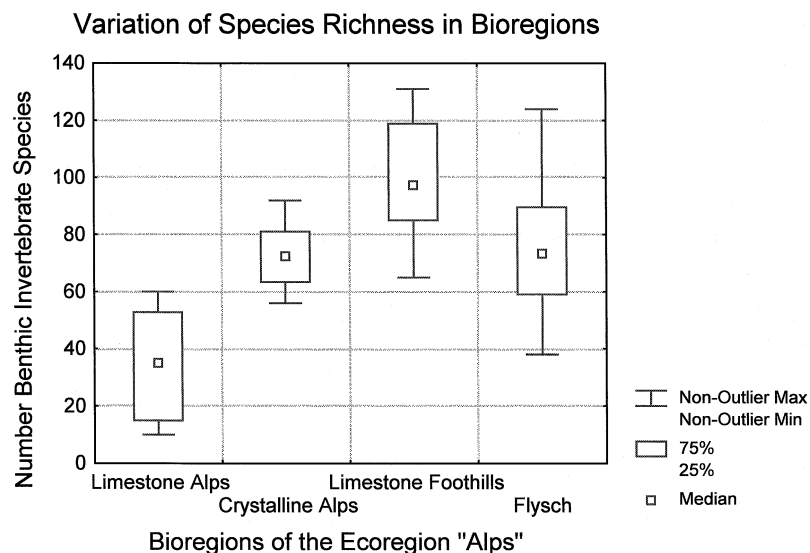


Figure 10. Variation of species richness in rivers in bioregions of the ecoregion 'Alps'.

Further analysis of sites in the Central Crystalline Alps (Fig. 6) gave evidence of three distinct bioregions: streams in glaciated areas, streams in non-glaciated areas and the streams of the lower ridges and foothills.

We continued this procedure of regional analysis of the biota, by subdivision of the regions according to the *a priori* hypotheses, and reanalysis of the biota of the subregions. *A priori* subregions that failed to show separation from other subregions in the ordination or clustering were not considered further. This analysis of the benthic invertebrate fauna of Austrian rivers led to a set of 15 bioregions that have been delineated for Austria (Table 1, Fig. 7), and are described in detail by Moog et al. (2001b).

Comparable analyses of the periphytic algae and the fish fauna confirmed our findings (unpublished reports), and the macroinvertebrate-based spatial typology of Austrian rivers was adopted as the 'Fluvial Bioregions of Austria' (Koller-Kreimel, 2002). In extension of the bioregions a couple of large rivers have been separated as 'special types'.

Validation of bioregions outside of Austria

Finally, we examined the validity of the Austrian bioregions with data from neighboring countries, also part of the AQEM EU Research project (Brabec et al., 2004; Lorenz et al., 2004). Bioregions at the Austrian frontier would be expected to straddle the border and occur on both sides. We analyzed reference and good

sites ($n = 108$; Hungarian Lowlands: 12, Alps: 46 and Central Highlands: 50 sites) from adjacent areas in Italy, Germany and the Czech Republic (Fig. 8). The scatter plot clearly shows the divisions among the Alps, Central Highlands and Hungarian Lowlands ecoregions. In addition, the alpine cluster shows two distinct groupings of sites within the crystalline and limestone Alps.

The results clearly indicated that ecoregions and bioregions respectively are robust across political boundaries. The bioregion concept thus fulfils the goal to be independent from political borders; and this concept can be used in Europe when implementing the WFD.

Bioregions and ecological indicators

One of the objectives of a regional classification is to improve the precision of biological indicators, such as the metrics used in a biotic index. As an example, we show the distribution of Trichoptera species richness, among four of the regions identified in the AQEM project (Fig. 9).

The box plots of Trichoptera species richness show a clear variation among the bioregions. Streams of the Hungarian Lowlands had the lowest Trichoptera richness (5–11 species), the two Alps bioregions had somewhat higher richness, and the Granite & Gneiss Region had the highest Trichoptera richness (15–18 species). The fact that Trichoptera species number (as one example) of rivers in the Crystalline Alps and

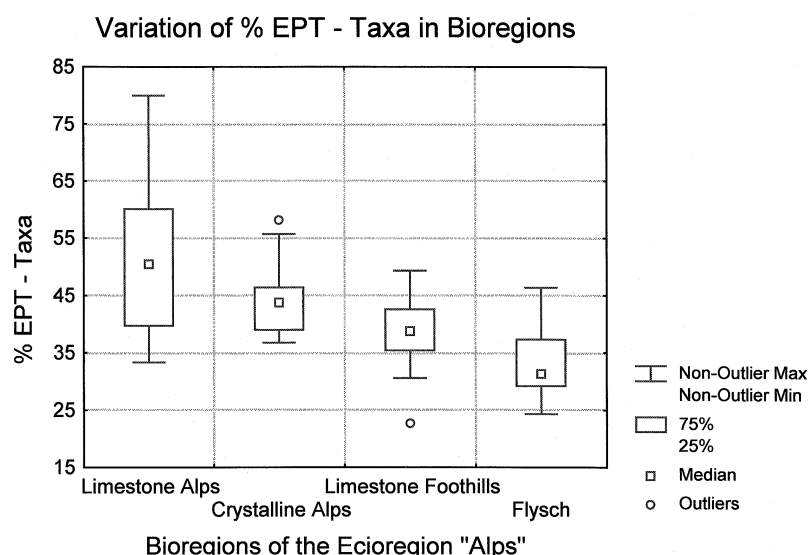


Figure 11. Variation of % EPT-taxa in rivers in bioregions of the ecoregion 'Alps'.

the Limestone Mountains are clearly different indicates that besides the ecoregion approach the bioregions also serve as strong predictors of metric ranges.

To demonstrate the distribution of metrics ranges among bioregions within one ecoregion, the following four bioregions in the ecoregion 'Alps' were examined (in order of decreasing altitude range: Limestone Alps (19 sampling sites; altitude 645–1717 m), Non-glaciated Crystalline Alps (16 sampling sites; altitude 870–1291 m), Limestone Foothills (18 sampling sites; altitude 400–807 m), Flysch (Sandstone) Mountains and Ridges (11 sampling sites; altitude 250–1425 m). Out of about 20 core metrics the following three metrics are presented in Figs 9–11 for example: species richness (number of species); % EPT species and share of collector-gatherers (within a total range from 1 to 10).

The box plots of species richness (number of benthic invertebrate species) vary significantly among the alpine bioregions (Fig. 10). From 10 to 130 species are found at single sites, with greater numbers of species at lower altitudes. Sites in the Limestone Alps are at high altitudes and are inhabited by an impoverished community compared to the sites in the Limestone Foothills. The large whiskers of the Flysch sites are probably a result of different river types due to the broad distribution of altitudes from 250 to 1425 m.

The metric % EPT species was high (above about 40%) at the higher altitudes and decreased at lower altitudes, with the lowest % of EPT taxa in the Flysch rivers.

The share of collector-gatherers was higher at lower altitudes, indicating a natural higher portion of fine particulate organic matter in those bioregions.

Discussion

We have used a systematic procedure of analyzing species composition among relatively large ecoregions, and then dividing the ecoregions into smaller geographic units, which, based on ecological knowledge and theory, were expected a priori to differ in their species composition. The subregions were based on considerations of altitude, geology, and river type. Those subregions that could be shown to differ substantially (in NMS ordination) from other subregions were retained as bioregions; subregions that did not differ substantially were recombined.

Resolution of the bioregions required a sufficient number of reference sites in each candidate region and homogeneous data quality within each analysis. At the broader ecoregion level, genus-level taxonomy was sufficient to delineate and characterize the four ecoregions examined. Finer spatial resolution of the smaller bioregions also required finer taxonomic resolution: to species or lowest practical level. The species-level resolution also required considerable effort to standardize the quality of data sets developed by multiple agencies or laboratories. The application of fine taxonomic resolution and high quality was rewarded with a finer geographic delineation than has been attained

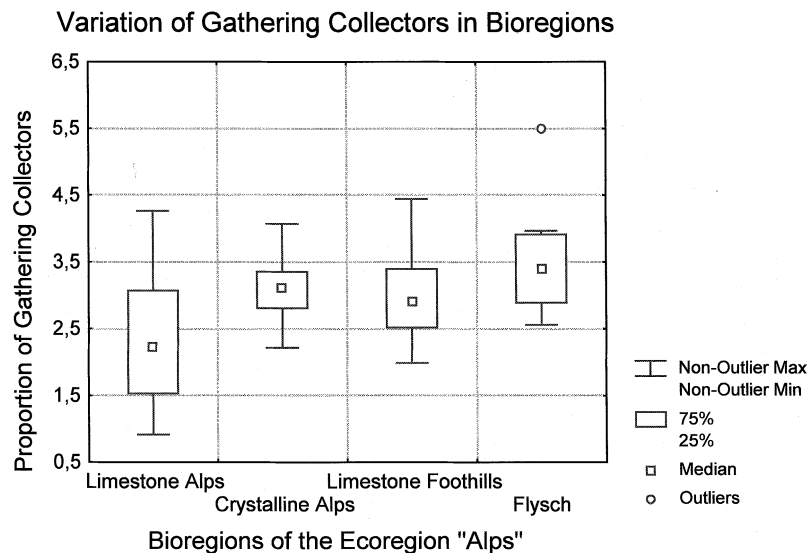


Figure 12. Variation of the proportion of collector-gatherers in rivers in bioregions of the ecoregion 'Alps'.

in other studies (e.g., Barbour et al., 1996; Gerritsen et al., 2000).

Figures 9–12 demonstrate that benthic assemblages are more homogeneous within a bioregion than among different bioregions. Within the bioregions the natural variety of several biological measures (metrics) is low. Consequently, for applied monitoring and river assessment, an effective classification improves the discriminatory power of measures for evaluating ecological status.

The ecoregion concept was confirmed as a valuable tool for the prediction of Austrian river faunas. Our analysis also showed that bioregions should serve as a typological basis for analysis of the benthic invertebrates to evaluate the ecological status of streams within the context of the WFD.

References

- Austrian Standards ÖNORM M 6232, 1997. Guidelines for the Ecological Survey and Evaluation of Flowing Surface Waters. Austrian Standards Institute, Vienna: 38 pp.
- Barbour, M. T., J. Gerritsen, G. E. Griffith, R. Frydenborg, E. McCarron, J. S. White & M. L. Bastian, 1996. A framework for biological criteria for Florida streams using benthic macroinvertebrates. *J. N. Am. Benthol. Soc.* 15: 185–211.
- Barbour, M. T., J. Gerritsen, B. D. Snyder & J. B. Stribling, 1999. Rapid Bioassessment Protocols for Use in Wadeable Streams and Rivers. Periphyton, Benthic Macroinvertebrates, and Fish, Second Edition. EPA 841-B-99-002. U.S. Environmental Protection Agency; Office of Water; Washington, D. C.
- Brabec, K., S. Zahrádková, D. Němecová, P. Pařil, J. Kokeš & J. Jarkovský, 2004. Assessment of organic pollution effect considering differences between lotic and lentic stream habitats. *Hydrobiologia* 516: 331–346.
- Chovanec, A., H. Heger, V. Koller-Kreimel, O. Moog, T. Spindler & H. Waidbacher, 1994. Anforderungen an die Erhebung und Beurteilung der ökologischen Funktionsfähigkeit von Fließgewässern – eine Diskussionsgrundlage. *ÖWAW* 46(11/12): 257–264.
- Chovanec, A., P. Jäger, M. Jungwirth, V. Koller-Kreimel, O. Moog, S. Muhar & S. Schmutz, 2000. The Austrian way of assessing the ecological integrity of running waters: A contribution to the EU Water Framework Directive. *Hydrobiologia* 422/423: 445–452.
- Davies, P. E., 2000. Development of a national river bioassessment system, AUSRIVAS in Australia. In Wright, J. F., D. W. Sutcliffe & M. T. Furse (eds.), *Assessing the biological quality of fresh waters – RIVPACS and other techniques*. *Freshwat. Biol.* 113–124.
- European Commission, 2000. Directive 2000/60/EC. Establishing a framework for community action in the field of water policy. European Commission PE-CONS 3639/1/100 Rev 1, Luxembourg.
- Fink, M., O. Moog & R. Wimmer, 2000. Fließgewässer – Naturräume Österreichs. UBA Monographien 128, Wien, 110 pp.
- Gerritsen, J., M. T. Barbour, & K. King, 2000. Apples, oranges, and ecoregions: on determining pattern in aquatic assemblages. *J. N. Am. Benthol. Soc.* 19(3): 487–496.
- Hawkins, C. P., R. H. Norris, J. N. Hogue, & J. Feminella, 2000. Development and evaluation of predictive models for measuring the biological integrity of streams. *Ecol. Appl.* 10: 1456–1477.
- Hemsley-Flint, B., 2000. Classification of the biological quality of rivers in England and Wales. In Wright, J. F., D. W. Sutcliffe & M. T. Furse (eds), 2000. *Assessing the biological quality of fresh waters – RIVPACS and other techniques*. *Freshwat. Biol.* 55–69.
- Hering, D., O. Moog, L. Sandin & P. F. M. Verdonchot, 2004. Overview and application of the AQEM assessment system. *Hydrobiologia* 516: 1–20.
- Hughes, R. M. & D. P. Larsen, 1988. Ecoregions: An approach to surface water protection. *J. Wat. Pollut. Cont. Fed.* 60: 486–493.
- Illies, J. (ed.), 1978. *Limnofauna Europaea*. 2., überarbeitete und ergänzte Auflage, G. Fischer Verlag, Stuttgart, New York, 552 pp.

- Lorenz, A., D. Hering, C. Feld & P. Rolauffs, 2004. A new method for assessing the impact of hydromorphological degradation on the macroinvertebrate fauna of five German stream types. *Hydrobiologia* 516: 107–127.
- Koller-Kreimel, V. (ed.), 2002. Typisierung der österreichischen Fließgewässer im Sinne der Vorgaben des Anhangs II der WRRL – Arbeitskreis Ökologie. Draft-Version from September 2002. Ministry of Agriculture, Forestry, Environment and Water Management: 12 pp.
- Koller-Kreimel, V. & A. Chovanec, 1999. Water Framework Directive – Ecological assessment of surface water status. International Conference on EU Water Management – Framework Directive and Danubian Countries. 21–23 June 1999, Bratislava, Proceedings: 93–110.
- Legendre, P. & L. Legendre, 1998. Numerical Ecology. Second English Edition. Elsevier Science, Amsterdam, xv + 853 pp.
- Logan, P., 2001. Ecological quality assessment of rivers and integrated catchment management in England and Wales. In Ravera, O. (ed.), Scientific and legal aspects of biological monitoring in Freshwater. *J. Limnol.* 60 (Suppl. 1): 25–32.
- Logan, P. & M. Furse, 2002. Preparing for the European Water Framework Directive – making the links between habitat and aquatic biota. *Aquatic Conserv.: Mar. Freshw. Ecosyst.* 12: 425–437.
- McCune, B. & M.J. Mefford, 1999. PC-ORD. Multivariate Analysis of Ecological Data, Version 4. MjM Software Design, Gleneden Beach, Oregon, USA: 237 pp.
- Moog, O., H. Nesemann, & T. Ofenböck, 2001a. Österreichs Anteil an den österreichischen Ökoregionen gemäß EU-Wasserrahmenrichtlinie – eine deduktive Analyse landschaftsprägender Milieufaktoren. *Österr. Wasser- und Abfallwirtschaft Wien*, 52 (7/8): 204–209.
- Moog, O., A. Schmidt-Kloiber, T. Ofenböck & J. Gerritsen, 2001b. Aquatische Ökoregionen und Bioregionen Österreichs – eine Gliederung nach geoökologischen Milieufaktoren und Makrozoobenthos-Zönosen. Bundesministerium für Land- und Forstwirtschaft, Umwelt und Wasserwirtschaft. Wasserwirtschaftskataster, Vienna.
- Moss, D., 2000. Evolution of statistical methods in RIVPACS. In Wright, J. F., D. W. Sutcliffe & M. T. Furse (eds), Assessing the biological quality of fresh waters – RIVPACS and other techniques. *Freshwat. Biol.*: 25–38.
- Moss, D., M. T. Furse, J. F. Wright, & P. D. Armitage, 1987. The prediction of the macro-invertebrate fauna of unpolluted running-water sites in Great Britain using environmental data. *Freshwat. Biol.* 17: 41–52.
- Nijboer, R. C., R. K. Johnson, P. F. M. Verdonshot, M. Sommerhäuser & A. Buffagni, 2004. Establishing reference conditions for European streams. *Hydrobiologia* 516: 91–105.
- Nijboer, R. C. & P. F. M. Verdonshot, 2000. Taxonomic adjustment affects data analysis: an often forgotten error. *Verh. int. Ver. Limnol.* 27: 1–4.
- Omernik, J. M., 1987. Ecoregions of the conterminous United States. *Ann. Assoc. Am. Geographers* 77(1): 118–125.
- Omernik, J. M., 1995. Ecoregions: A framework for managing ecosystems. *The George Wright Forum* 12(1): 35–51.
- Reynoldson, T. B., R. C. Bailey, K. E. Day & R. H. Norris, 1995. Biological guidelines for freshwater sediment based on Benthic Assessment of Sediment (the BEAST) using a multivariate approach for predicting biological state. *Aust. J. Ecol.* 20: 198–219.
- Schmidt-Kloiber, A., O. Moog, & J. Gerritsen, 2001. Die aquatischen Ökoregionen Österreichs – Ergebnisse multivariater Analysen von Makrozoobenthos-Zönosen. *Österreichs Fischerei*, 54/2001: 154–161.
- Schmidt-Kloiber, A., 2002. Deduktion der Fließgewässer-Bioregionen auf Basis multivariater Analysen der wirbellosen Bodenfauna. PhD Thesis, University of Agricultural Sciences Vienna, 150 pp.
- Schmidt-Kloiber, A. & R. C. Nijboer, 2004. The effect of taxonomic resolution on the assessment of ecological water quality classes. *Hydrobiologia* 516: 269–283.
- USEPA (U.S. Environmental Protection Agency), 1996. Biological criteria: Technical guidance for streams and small rivers. U.S. Environmental Protection Agency, Office of Water, Washington, DC. EPA-822-B96-001.
- Van Sickle, J., 1997. Using mean similarity dendrograms to evaluate classifications. *JABES* 2: 370–388.
- Van Sickle, J. & R. M. Hughes, 2000. Classification strength of ecoregions, catchments, and geographic clusters for aquatic vertebrates in Oregon. *J. N. Am. Benthol. Soc.* 19(3): 370–384.
- Verdonshot, P. F. M. & R. C. Nijboer, 2004. Testing the European stream typology of the Water Framework Directive for macroinvertebrates. *Hydrobiologia* 516: 35–54.
- Wimmer, R. & A. Chovanec, 2000. Fließgewässertypen in Österreich als Grundlage für die Erarbeitung eines Überwachungsnetzes im Sinne des Anhangs II der EU Wasserrahmenrichtlinie. Bundesministerium f. Land- u. Forstwirtschaft, Umwelt u. Wasserwirtschaft, Wasserwirtschaftskataster, 37 pp.
- Wimmer, R., A. Chovanec, O. Moog, M. H. Fink & D. Gruber, 2000. Abiotic Stream Classification as a Basis for a Surveillance Monitoring network in Austria in Accordance with the EU Water Framework Directive. *Acta hydrochim. hydrobiol.* 28(4): 177–184.